1st lecture

Monday, August 24, 2026 9:49 AM

Math 6660 - FEMs

Context:

5610: interp, int, ODEs

5620: Numerical Lin. Algeb

6600 - FEMs

6670 → FVMs

5170 → FDMs



---

why FEM?

Original problem is continuous

(i) Theoritical:

FDM: approximate the solution to a discrete number of points;

FEM: keep orginal continous problem (PDE), but replace (approx) function

spaces.

All theory of PDE carried to FEM

② practical reasons:

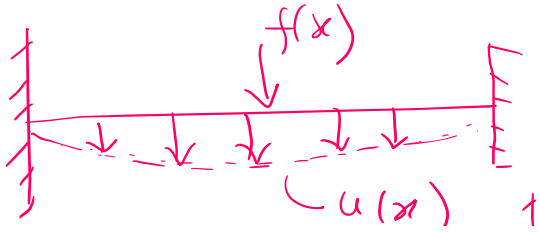
→ can be formulated ~~very~~ on unstructured grid

→ lots of flexibility in choice of function spaces

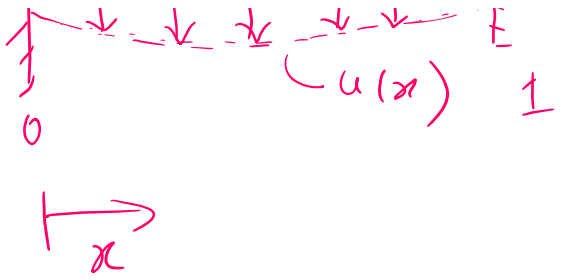
→

③ Note: much of theory of FEMs, is based on elliptic PDE theory. Parabolic PDE are ok. Hyperbolic PDEs are more challenging

Two point BVP ? (1D version of poission)



$\rightarrow -\left(a(x)\,u'\right)' \equiv f(x)$



$\rightarrow -(a(x)u')' \equiv f(x)$

$0 < x < 1$

$u(0) = u(1) = 1$

$u :=$ "displacment"
$f :=$ "external load" } given
$a =$ "modulus of elasticity" } given

Assume: $0 < a_1 \leq a(x) \leq a_2 < \infty$

$\forall \rightarrow x \in [0,1]$

Note $\mathcal{L}(u) = -(au')'$ with $u(0) = u(1) = 0$

(self adjoint operator) ~~is~~ a linear differential operator ($\mathcal{L}(u)$ is self-adjoint

$$\int_0^1 v\mathcal{L}(u)\,dx = \int_0^1 v(au')'\,dx$$



$$= \int_0^1 v' a u' dx - \left[v a u' \right]_{x=0}^{x=1}$$

assume $u(0) = u(1) = 0$ and $v(0) = v(1) = 0$

$$= \int_0^1 v' a u' dx$$

$$= -\int_0^1 u(a'v') dx + \left[u a v' \right]_{x=0}^{x=1}$$

$$= -\int_0^1 u (av)' dx = \int_0^1 u \, \mathcal{J}(v) dx$$

$$\therefore \int_0^1 v \, \mathcal{J}(u) dx = \int_0^1 u \, \mathcal{J}(v) dx$$

(v: test function)

This is ~~not~~ what it means by to be self-adjoint

Note: self adjoint differential operator $\mathcal{J}$

Note: self adjoint differential operator $\mathcal{L}$ is analogous to symmetric matrices

Note: $\mathcal{L}(w) = 0$, $w(0) = w(1) = 0$

$\Leftrightarrow$ only solution that satisfy $\mathcal{L}$ & B.C.

$w = 0$

$\therefore$ if $\mathcal{L}$ is "invertible" $\Rightarrow$ for a given f there exist a unique u and f.

Note: if $0 < a < \alpha$ and f are sufficiently smooth

need: $a \in C^1(0,1)$ and $f \in C(0,1)$

then $u \in C^2(0,1)$

then, $u \in C^2(0,1)$

Variational Formulation:

Variational Formulation:

Multiply our BVP by a test function and integrate over the domain $[0, 1]$

$$-\int (au')' v \, dx = \int_0^1 f v \, dx$$

(f: force)

Then integrate by parts

$$\int_0^1 au'v' \, dx - [au'v]_0^1 = \int_0^1 f v \, dx$$

assume, v satisfy BCs:

$$v(0) = v(1) = 0$$

In particular, define solution space,

$$V = \left\{ v : \int_0^1 v^2 dx < \infty \; ; \int_0^1 [v']^2 dx < \infty, \right.$$

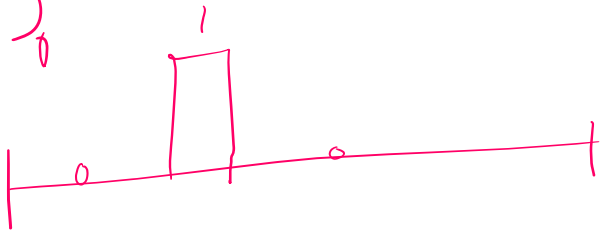
$$\left. v(0) = v(1) = 0 \right\}$$

Note:

f ... v are continous

functions in V are continous

(1) $\int_0^1 v^2 dx < \infty$ admits discontinous :



(2) $\int_0^1 [v']^2 dx$ does not allow discontinous in V.

2nd lecture

2 point BVP

$-(au')' = f$ on fixed domain $0 < x < 1$

$u(0) = u(1) = 0$

find $u \in C^2(0,1)$

---

variational formulation

$V = \{v : \int_0^1 v^2 dx < 0 \longrightarrow$ defining a space

$v(0) = v(1) = 0\}$

$$\int_0^1 (v')^2 dx < \alpha, \quad v(0) = v(1) = 0 \Big\}$$

Note: functions on V are continuous ($v \in C^0(0,1)$)

ex of discontinuous

$$H(x) = \begin{cases} 1 & \frac{1}{4} < x < \frac{3}{4} \\ 0 & \text{otherwise} \end{cases}$$

↳ not continuous

$H \notin C^0(0,1)$

but

$H \in C^{-1}(0,1)$

↳ finite jump

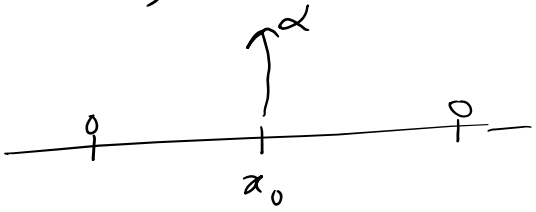
So,

$$\int_0^1 H^2 dx = \int_{1/4}^{3/4} dx = \frac{1}{2}$$

Now, the derivative,

$$H'(x) = \delta\left(x - \tfrac{1}{4}\right) - \delta\left(x - \tfrac{3}{4}\right)$$

$\delta(x - x_0)$:



δ property

$$\int_{x_0-\epsilon_1}^{x_0+\epsilon_2} \delta(x-x_0)\,dx = 1$$

for any $\epsilon_1, \epsilon_2 > 0$

Another one

$$\int_{x_0-\epsilon_1}^{x_0+\epsilon_2} f(x)\,\delta(x-x_0)\,dx = f(x_0)$$

f ~ continuous function

Now,

$$\int_0^1 (H')^2\,dx = \int_0^1 \left(\delta_{1/4} - \delta_{3/4}\right)^2 dx$$

$$= \int_0^1 \left(\delta_{1/4}^2 - 2\delta_{1/4}\delta_{3/4} + \delta_{3/4}^2\right) dx$$

Here, $\Rightarrow \int_0^1 \delta\left(x-\frac{1}{4}\right)\delta\left(x-\frac{1}{4}\right)dx = +\infty$

$\therefore H(x) \notin V$ because H' is not square

integrateable

Variational problem:

Find $u \in V$ such that

$$\int_0^1 a u' v' \, dx = \int_0^1 f v \, dx$$

for all $v \in \hat{V}$.

Note: u is the trial function

v are the test function

In this case u & v comes from the same function space $\therefore V$

Q: How are (2 pt BVP) and (Vari problem) related?

→ BVP problem require $u \in C^2(0,1)$

→ Variational problem only need $u \in V$

→ Variational problem only need $u \in V$

claim 1:

If variational problem has a solution it is unique.

proof: (by contradiction)

assume ∃ 2 different solu: $u \neq U$

$$\int_0^1 a u' v' \, dx = \int_0^1 f v \, dx \quad \Big| \quad \int_0^1 a U' v' dx = \int_0^1 f v dx$$

Then, subtract:

$$\Rightarrow \int_0^1 a (u-U)' v' dx = 0 \quad \text{take, } v = u - U \in V$$

Now

$$\int_0^1 a \underbrace{\left((u-U)' \right)}_{v} dx = 0$$

$$\Downarrow$$

$$(u-U)' = 0 \quad \text{on } (0,1)$$

$$\Downarrow\Downarrow$$

V

$u = U + c$

but, by BCs, $\therefore c = 0 \Rightarrow u = U$ (proved)

Claim 2 Solu to the BVP is a solution to variational problem

Proof: BVP has unique solution ~~as~~ is $u \in C^2(0,1)$

and $u(0) = u(1) = 0$

If $u \in C^2(0,1)$ + BCs $\Rightarrow u \in V$

Then BVP & variational problem are equivalent via int-by-parts

Claim 3: If $f \in C^0(0,1)$, and $a \in C^1(0,1)$, and

Lemma: If

$a \in C^1(0,1)$, and

$0 < a_1 \leq a(x) \leq a_2 < \infty$

$\forall \quad x \in (0,1)$

Then solution to variational problem is a solution ~~to~~ to BVP.

proof:

assume for the moment that $u \in C^2(0,1)$

$$\int_0^1 \left(a u' v' - f v \right) dx = 0$$

reverse int-by-parts:

$$\int_0^1 \underbrace{\left[(au')' + f \right]}_{S(x)} v \, dx = 0$$

Goal: to show $S(x) = 0$ for all $x \in (0,1)$

(a) Assume $S(\xi) = [(au')' + f]\Big|_{x=\xi} > 0$

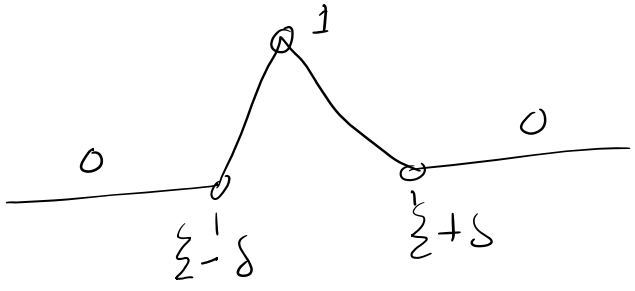
(proof by contradiction)

at some point $\xi \in (0,1)$

Note: that $S(x)$ is continuous in x.

By continuity: $S(x) > 0$ for $x \in [\xi - \delta, \xi + \delta]$ for some $\delta > 0$

choose v to be the following "hat" function



but $\int_0^1 S(x)\, v(x)\, dx > 0$

which is a contradiction.

$\therefore$ $S(x)$ cannot be positive in $(0,1)$

b) Redo this argument w/ $S(\xi) < 0$

b) Recall this $\Rightarrow$ $S(x)$ cannot be negative in $(0,1)$

When can we expect $u \in C^2(0,1)$?

(if $f \in C^0(0,1)$, $a \in C^1(0,1)$ and

$*$ $0 < a_1 \le a(x) \le a_2 < \infty$ $\forall x \in (0,1)$)

Note:

(1) If $*$ is satisfied: $\rightarrow$ u is a strong soln of 2-pt BVP $\equiv$ (variational problem)

(2) If $*$ is not satisfied,

2-pt BVP will not have a solu.

but variational problem may still have a solution. $\rightarrow$ u (if exist) is a weak soln of (2-pt BVP)

3rrd

Thursday, August 27, 2026 1:02 PM

2 pt BVP

$$\left.\begin{aligned} -(a u')' &= f \quad \text{on} \quad (0,1) \\ u(0) &= u(1) = 0 \end{aligned}\right\}$$

$f \in C^0$, $a \in C^1$, $0 < a_1 < a(x) < a_2 < \infty$

$\forall\ x \in [0,1]$

variational problem

Find $u \in V = \left\{ v : \int v^2 dx < \infty, \right.$

$\left. \int (v')^2 dx < \infty \right.$

$v(0) = v(1) = 0$

such that

$$\int_0^1 a u' v' \, dx = \int_0^1 f v \, dx$$

$\forall\ v \in V$

"Energy" minimization problem:

(not total energy)

Define the following energy functional:

$$F(w) = \frac{1}{2} \int_0^1 a (w')^2 \, dx - \int_0^1 f w \, dx$$

$\int_0^1 a(w)\,dx$ — $\int_0^1$

w is a function — internal Energy — potential Energy

kinetic Energy — potential Energy

Min problem: Find $u \in V$ such that

$F(u) \leq F(w) \quad \forall\, w \in V$

claim: Min. Problem and variational problem are equivalent.

proof:

(1) (variational problem) $\Rightarrow$ (Min Prob):

Assume $u \in V$ satisfies:

$$\int_0^1 a u' v' dx = \int_0^1 f v dx$$

$$\forall \quad v \in V$$

Let $w \in V$, s.t. $w = u + v$

$$F(w) = f(u+v)$$

$$= \frac{1}{2}\int_0^1 a(u'+v')^2 dx - \int_0^1 fu\,dx - \int_0^1 fv\,dx$$

$$= \frac{1}{2}\int_0^1 a(u')^2 dx + \int_0^1 a u'v'\,dx + \frac{1}{2}\int_0^1 a(v')^2 dx$$

$$- \int_0^1 fu\,dx - \int_0^1 fv\,dx$$

variational problem

$$= F(u) + \frac{1}{2}\int_0^1 a(v')^2 dx$$

$$\Rightarrow F(w) = F(u) + \frac{1}{2}\int_0^1 a(v')^2 dx$$

$\therefore F(w) > F(u)$ (proved)

$\forall\ w \in V$

(2) claim: min prob $\Rightarrow$ Var Problem

Assume, $u \in V$ satisfies

$$F(u) \leq F(w) \quad \forall\, w \in V$$

let, $g(\epsilon) = F(u + \epsilon v)$ where, $v \in V$ is arbitrary but fixed

$$g(\epsilon) = \frac{1}{2}\int_0^1 a(u' + \epsilon v')^2 dx - \int_0^1 f(u + \epsilon v) dx$$

$$= \frac{1}{2}\int_0^1 a(u')^2 dx + \epsilon \int_0^1 a u' v' dx + \frac{1}{2}\epsilon^2 \int_0^1 a(v')^2 dx$$

$$- \int_0^1 f u dx - \epsilon \int_0^1 f v dx$$

$g(\epsilon)$ is a quadratic poly in ϵ with a positive quad term.

the unique global minimizer happens at

$\epsilon = 0$ and $g'(\epsilon = 0) = 0$

$$g'(\epsilon) = \int_0^1 u' v' dx - \int_0^1 f v dx \quad \epsilon \int_0^1 a(v')^2 dx$$

$$g'(\epsilon=0) = \int_0^1 a u'v' \, dx - \int_0^1 f v \, dx = 0$$

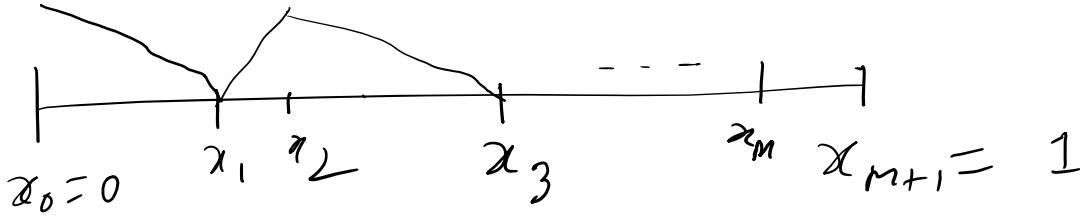
which is exactly the variational problem

variational problem is completely equivalent of this energy minimization. ▨

// Our 1st Finite element method:

Idea: Solve (variational problem) or (min problem) approximating by replacing V by a finite dimensional subspace V_h

let T_h be mesh on $(0,1)$



let, $h_j = x_j - x_{j-1}$

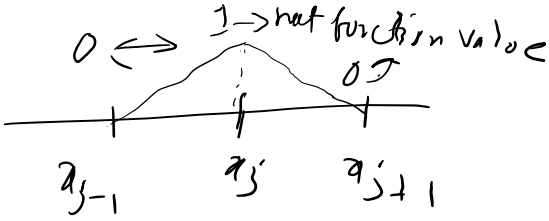
for $j = 1, 2, \ldots m+1$

Simplest $V_h \subset V$: Space of continuous piecewise linear polynomial on T_h

This is called the CG(1) method.

Continuous Galerkin method w poly of degree 1

Any function in V_h can be uniquely written as follows



$$U(x) = \sum_{i=1}^{m} \xi_j \phi_j(x)$$

ξ_j : coefficient

and $\phi_j(x)$ is a hat function associated with x_j.

$$\phi_j(x) = \begin{cases} \frac{x - x_{j-1}}{h_j} & x_{j-1} \leq x < x_j \end{cases}$$

$\psi_j(x) = \begin{cases} h_j & \\ \dfrac{x_{j+1} - x}{h_{j+1}} & x_j \le x \le x_{j+1} \end{cases}$

everything else out of limit is zero